

An exploration of the relation between the visual attributes of thumbnails and the view-through of videos: The case of branded video content

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ABSTRACT

While browsing through an online video platform, potential viewers decide which videos to click and watch based on the information and impression they obtain from thumbnails. Therefore, a thumbnail needs to be able to tell potential viewers what the video is about (i.e., be informative), and at the same time, a thumbnail needs to grab potential viewers' attention (i.e., be visually appealing). Drawing from the elaboration likelihood model (ELM), we take the visual attributes that are relevant to the informativeness of thumbnails (e.g., element complexity and object complexity) as central route cues, take the visual attributes that are relevant to the aesthetics of thumbnails (e.g., celebrity endorsement, colorfulness, brightness, and image quality) as peripheral route cues, and explore the relation between these visual attributes of thumbnails and the view-through of videos using 3,745 marketing videos posted by 38 top brands across 4 industries – the automobile, beverage, restaurant, and toy industries – on YouTube. We construct our variables by extracting features from thumbnails using image mining techniques. Our study contributes to the literature on information systems and marketing by enhancing our understanding of the role of the visual attributes of images in persuading viewers in the online environment. Furthermore, our study provides a possible theoretical basis for studies on information processing and decision support that focus on automatic thumbnail selection. Finally, our study provides useful and practical guidelines for designing templates for optimal thumbnails that grab potential viewers' attention and yield more video views.

1. Introduction

In recent years, the online video platform market has grown substantially. As of May 2019, YouTube, one of the most popular online video platforms, had more than 2 billion monthly unique users worldwide [78]. These users uploaded more than 500 hours of video every minute [30] and watched more than 1 billion hours of video each day [59]. It is predicted that video traffic will account for 82% of all internet traffic by 2022 [16]. Furthermore, consumers say that they prefer video content over any other form of content [55]. They discover brands and products through videos [71] and engage with brands and make purchasing decisions after watching videos [9]. Accordingly, online video platforms have become a prominent marketing channel for businesses to reach a wider range of target customers and engage with them. In 2019, 85% of businesses used video content as a marketing tool [34]. Approximately 24% of those businesses uploaded video content weekly, and approximately 13% uploaded video content daily [11]. In 2018, advertisers spent \$90.7 billion on video marketing, and this value is

expected to increase to \$102.8 billion by 2023 [46]. Eighty-eight percent of advertisers said that they were satisfied with the return on investment (ROI) of their video marketing efforts in 2018 [62]. However, with more businesses becoming aware of the advantages of video marketing, the competition is becoming fierce.

According to an analysis by the Pew Research Center, 43,770 YouTube channels that had more than 250,000 subscribers as of January 2019 collectively uploaded 48,486 hours of video content in the first week of 2019. Those videos were viewed more than 14.2 billion times during the same time period; however, 10% of the videos were responsible for 79% of all those views [42]. Similarly, Chowdhury and Makaroff [15] show that among 47,771 YouTube videos in their sample, only 0.08% received more than 100,000 views. Approximately half of those videos were viewed less than 100 times, and a quarter of them were not even viewed 10 times. For video content, it is becoming more challenging to stand out from the crowd.

Industry experts have highlighted the importance of thumbnails for the view-through of videos [48]. Viewers decide which videos to click

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and watch when browsing through an online video platform based on the information and impression they obtain from thumbnails. A thumbnail is a frame taken from a video or a customized image that represents the content of the video. YouTube reports that although it provides a few thumbnail choices from the video itself based on some algorithms, 90% of top-performing videos use customized thumbnails rather than the “suggested” choices [26]. Anecdotal evidence suggests that “optimized” thumbnails increase the view-through of videos by 15% [81]. However, academia has paid little attention to investigating what aspects of thumbnails are associated with the view-through of videos.

A thumbnail is an abstract of a video; hence, it may need to be informative and tell potential viewers what the video is about. At the same time, a thumbnail needs to be visually appealing and grab potential viewers' attention. Drawing from the elaboration likelihood model (ELM) [65,66], we take the visual attributes that are relevant to the informativeness of thumbnails (e.g., the element complexity and object complexity of thumbnails) as central route cues, take the visual attributes that are relevant to the aesthetics of thumbnails (e.g., the celebrity endorsement, colorfulness, brightness, and image quality of thumbnails) as peripheral route cues, and explore the relation between these visual attributes of thumbnails and the view-through of videos.

Using data collected from 3,745 marketing videos posted by 38 top brands across 4 industries – the automobile, beverage, restaurant, and toy industries – on YouTube, we show that these visual attributes of thumbnails have statistically significant relations with the view-through of videos, which may suggest that these visual attributes persuade potential viewers to click and watch videos via the central route and peripheral route. In particular, having more types of design elements (e.g., product, model, text) in a thumbnail may help convey information; however, focusing on too many objects can hinder the processing of the information. Furthermore, thumbnails with a celebrity picture, with a combination of high colorfulness and high brightness or low colorfulness and low brightness, and with moderate image quality may be more effective in persuading potential viewers to click and watch videos via the peripheral route. The industry-level analysis reveals that potential viewers' reactions to the aesthetics of thumbnails may be different across different industries.

Our study makes the following contributions. First, a relatively small number of studies on information systems and marketing have explored the role of visual attributes of images in persuading viewers in the online environment perhaps because generating features from a large number of images was not easily feasible. Using recent deep learning-based image processing approaches, Shin et al. [76] explore the relation between the visual attributes of images and user engagement in online social media, and Zhang et al. [88] explore the relation between the visual attributes of images and the demand for Airbnb properties. Complementing these studies, our study provides additional insights and enhances our understanding of the role of the visual attributes of images in conveying information and in drawing viewers' attention in the context of branded video content on an online video platform. We use modern image processing techniques to extract features from thumbnails and construct visual attributes. Second, complementing a growing body of literature that studies the factors that affect the view-through of videos [6,43,49,58,85,89], our study explores the visual attributes of thumbnails as additional factors that may be associated with the view-through of videos. A vast majority of studies on thumbnails in image processing and decision support have focused on automatic thumbnail selection [12,28]. Our study provides a possible theoretical ground for future research in this stream of research. Finally, our study provides useful and practical guidelines for designing templates for “optimal” thumbnails that grab potential viewers' attention and yield more video views.

2. Context and relevant literature

2.1. Context: Branded video content

Online video platforms offer various types of advertisement options, including in-stream ads, bumper ads, overlay ads, and banner ads. Among them, we chose *branded video content* as the context to explore the relation between the visual attributes of thumbnails and the view-through of videos. Branded video content refers to videos sponsored or created by brands [73]. In particular, we focus on videos uploaded by brands on their YouTube brand channels (See Fig. 1). In general, branded video content is funny, emotional, heartwarming, or educational rather than promotional; however, consumers do not see much difference between branded video content and video advertisements [29]. Interbrand's top 100 brands, such as Disney, Coca-Cola, and Apple, had an average of 2.4 channels on YouTube and collectively uploaded a video every 18.5 minutes in 2015 [82]. LEGO had the most popular YouTube brand channel as of December 2019, with more than 8 million subscribers, more than 20,000 videos, and more than 10 billion views.¹

2.2. Relevant literature

There is a growing body of literature exploring the factors that affect the view-through of videos. These studies suggest that the view-through of videos is affected by the social capital of uploaders [6,43] and the characteristics of the videos, such as age [43], length [72], and viewer reviews [43]. Furthermore, Dean [20] shows that the number of views of a YouTube video is highly correlated with its ranking, which in turn is correlated with the size of the channel and the number of Likes that the video received. Complementing these earlier studies, our study focuses on the visual attributes of thumbnails as additional factors that may be associated with the view-through of videos.

The vast majority of the literature on thumbnails focuses on automatic thumbnail selection [12,28], but a few studies have explored the role of thumbnails in video views. In the mobile environment, Hürst et al. [37] examine the relation between the retrieval of videos and the characteristics of thumbnails, such as size and motion type, and Yoon and Kim [87] examine the relation between the selection of videos and the valence and arousal of thumbnails. Fu [25] explores whether the presence of thumbnails moderates the bandwagon effects of video views.

3. Theoretical background and conceptual framework

3.1. Theoretical background: Elaboration likelihood model

The ELM proposes that there are two routes to persuasion: the central route and the peripheral route [65,66]. When individuals are exposed to a persuasive message, depending on their elaboration likelihood, they process the message via the central route, the peripheral route, or both [44,67]. In the central route, individuals rely more on the content of the message, while in the peripheral route, they rely more on the appearance of the message.

The validity of the ELM has been verified in extensive studies [4]. Using the ELM as a theoretical basis, studies in information systems have explored the persuasiveness of elements of messages in the form of text, images, or videos in various contexts, including web personalization [36,79], website design [19], social media [76], and customer reviews [14]. Using the ELM as a theoretical framework, our study explores the persuasiveness of visual attributes of thumbnails in the context of branded video content.

Thumbnails are usually the first thing that potential viewers see when they browse through an online video platform. Hence, a

¹ Statistics are available from: <https://socialblade.com/youtube/user/lego>.

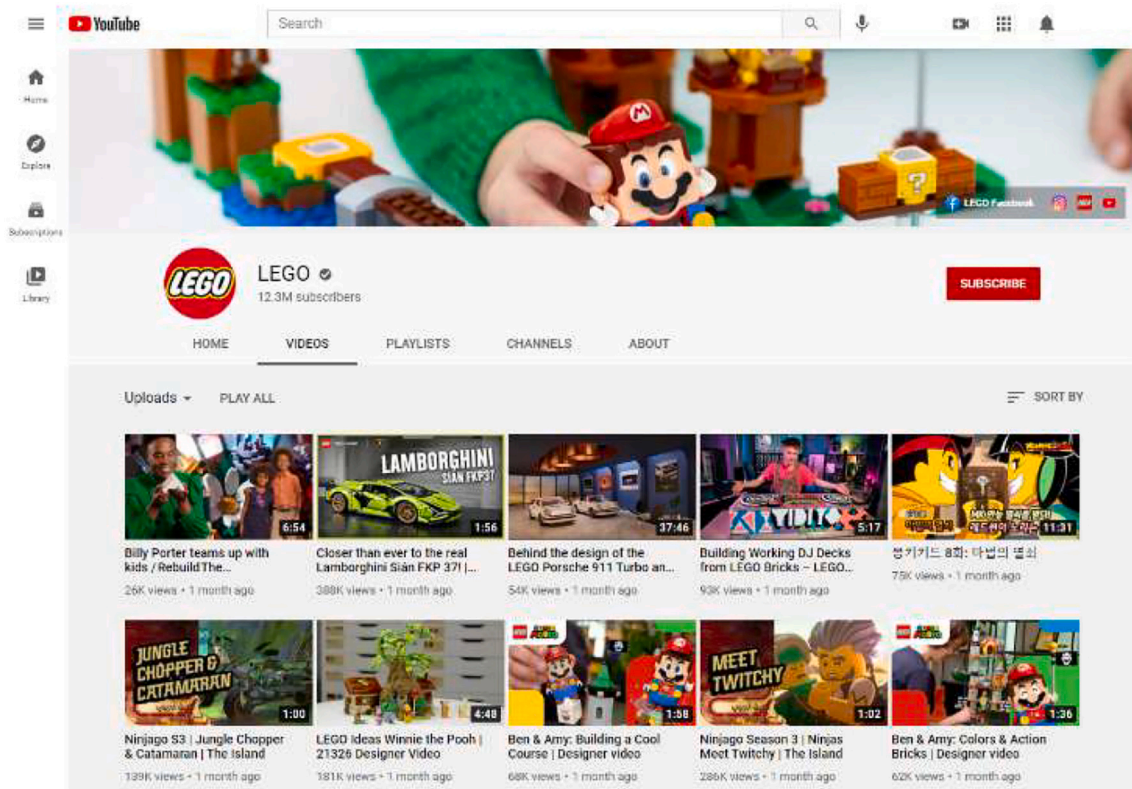


Fig. 1. Videos on the YouTube LEGO channel.

thumbnail needs to be persuasive in the sense that it needs to be able to tell potential viewers what the video is about (i.e., be informative), and at the same time, it needs to grab potential viewers' attention (i.e., be visually appealing) [10,26]. Drawing from the ELM framework, we take the visual attributes that are relevant to the *informativeness* of thumbnails as central route cues and the visual attributes that are relevant to the *aesthetics* of thumbnails as peripheral route cues.

3.2. Central route cues: Visual attributes relevant to the informativeness of thumbnails

The literature documents that the visual complexity of advertisement images is an important central route cue [76]. In the central route, advertisement images with higher visual complexity convey more information [41,52] but may hinder viewers in locating specific information [22,70]. Additionally, more visually complex designs are more engaging [47,63,68]; however, they can hurt communication effectiveness [2,7,27]. Taking these findings together, we posit that the *visual complexity* of thumbnails is a visual attribute that may persuade potential viewers to click and watch videos via the central route.

In the context of webpage design, Deng and Poole [21] suggest that visual complexity is composed of two dimensions: visual diversity, which measures the number of types of elements (e.g., text, graphic, links) present on a webpage, and visual richness, which measures the number of elements present on a webpage. Following this conceptualization of visual complexity, we posit that the visual complexity of thumbnails consists of two types of complexity: *element complexity*, which refers to the number of types of design elements used in a thumbnail, and *object complexity*, which refers to the number of objects included (focused on) in a thumbnail.

3.2.1. Element complexity

Like many visual advertisements, thumbnails convey information through two types of design elements: *text* and *picture* [13,40,50,77],

with two types of pictures: *product* and *model* [5]. Each of the thumbnails shown in Fig. 2a uses one of these three types of design elements (product vs. model vs. text). These different types of design elements are sometimes used together in one thumbnail, as depicted in Fig. 2b and c. Fig. 2b and c show thumbnails that use a combination of two types of design elements and a thumbnail that uses all three types of design elements, respectively. We claim that thumbnails that use more types of design elements are more element complex.

3.2.2. Object complexity

Following earlier studies [21,68], we claim that thumbnails that include (or focus on) more objects are more object complex. For instance, both thumbnails (a) and (b) shown in Fig. 3 use only one type of design element, product, and their element complexities are identical; however, thumbnail (b) is more object complex than thumbnail (a).

3.3. Peripheral route cues: Visual attributes that are relevant to the aesthetics of thumbnails

The visual attributes extracted from advertisement images that are peripheral route cues include the *celebrity endorsement*, *colorfulness*, *brightness*, and *image quality* [76].

3.3.1. Celebrity endorsement

Based on the findings of earlier studies showing that celebrity endorsement increases the attention paid to advertisements [1,23,45], we posit that thumbnails that include a celebrity picture are more visually appealing than thumbnails that do not include a celebrity picture. That is, the celebrity endorsement of thumbnails is a visual attribute that may persuade potential viewers to click and watch videos via the peripheral route.

3.3.2. Colorfulness

We refer to colorfulness as the variation in color within a thumbnail

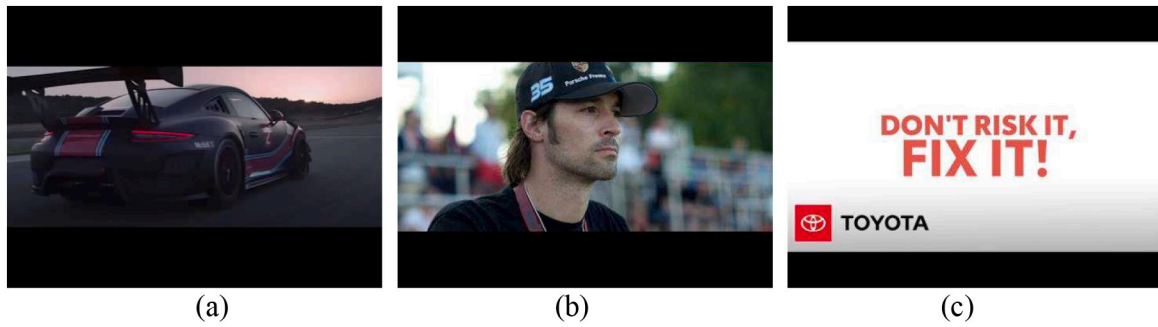


Fig. 2a. Thumbnails that use (a) a product design element, (b) a model design element, and (c) a text design element.

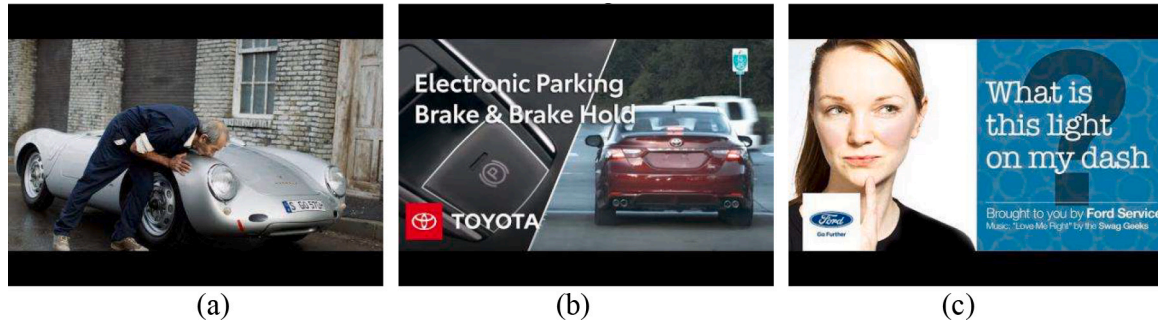


Fig. 2b. Thumbnails that use a combination of (a) product and model design elements, (b) product and text design elements, and (c) model and text design elements.



Fig. 2c. A thumbnail that uses all three types of design elements.

and claim that thumbnails with greater variations in color are more colorful than thumbnails with fewer colors. Colorfulness has been found to be one of the critical properties influencing the aesthetics of

designs [32,60,69]. However, the findings are mixed. Some studies have found that designs with high colorfulness are more appealing [69,83], whereas other studies have shown that designs with moderate colorfulness are more appealing [69,84]. Taking these mixed findings together, we posit that the colorfulness of thumbnails is a visual attribute that may persuade potential viewers to click and watch videos via the peripheral route.

3.3.3. Brightness

Brightness is considered another important color-related property affecting the aesthetics of designs [64,74]. Prior studies have shown that websites with brighter backgrounds receive higher ratings [51] and that mobile apps with brighter icon designs are downloaded more [83]. Accordingly, we posit that the brightness of thumbnails is a visual attribute that may persuade potential viewers to click and watch videos via the peripheral route.

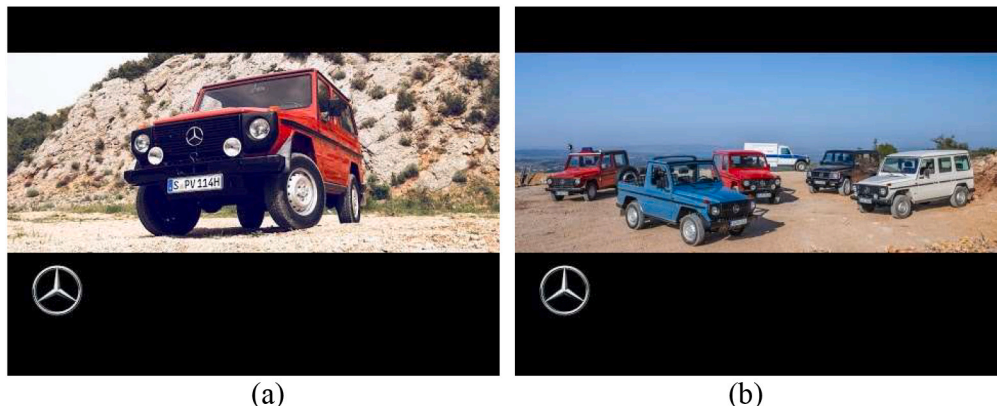


Fig. 3. Thumbnails with (a) low object complexity and (b) high object complexity.

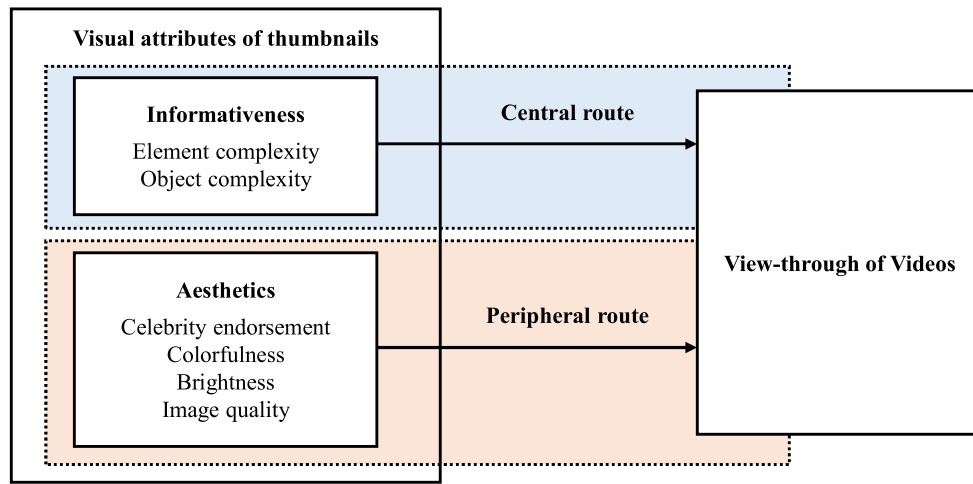


Fig. 4. Conceptual framework.

Table 1

The list of brands and the number of videos in our sample for each industry.

Industry	List of brands	Number of videos
Automobile	Mercedes-Benz, Toyota, BMW, Volkswagen, Honda, Nissan, Porsche, Ford, Audi	1,314
Beverage	Coca-Cola, Pepsi, Red Bull, Nescafé, Gatorade, Sprite, Monster, Mountain Dew, Dr. Pepper, Tropicana	429
Restaurant	Starbucks, McDonald's, Subway, KFC, Tim Hortons, Domino's Pizza, Burger King, Pizza Hut, Dunkin' Donuts, Chipotle	130
Toy	LEGO, Bandai Namco, Fisher-Price, Barbie, Nerf, Hasbro, Hot Wheels, My Little Pony, Mattel	1,870

3.3.4. Image quality

The literature has shown that advertisements with high-quality images are more effective than advertisements with low-quality images [8,75]. The digital retouching (or photoshopping) of advertisement images has naturally become a common tactic in advertising [53]. However, some recent studies have shown that retouching advertisement images does not necessarily improve the effectiveness of advertisements [17,80,88]. Over-retouched advertisement images can mislead consumers and ultimately lead to a negative attitude toward advertisements. Taking these equivocal findings together, we posit that the image quality of thumbnails is a visual attribute that may persuade potential viewers to click and watch videos via the peripheral route.

3.4. Conceptual framework

In summary, drawing from the ELM framework and taking the element complexity and object complexity of thumbnails as central route cues and the celebrity endorsement, colorfulness, brightness, and image quality of thumbnails as peripheral route cues, we explore the relation between these visual attributes of thumbnails and the view-through of videos. Our conceptual framework is summarized in Fig. 4.

4. Data and variables

4.1. Data

We collected data on branded video content posted in 2018 from 38 YouTube brand channels across 4 industries – the automobile, beverage, restaurant, and toy industries – using YouTube Data API v3. We chose the brands and industries included in our sample by using the following

procedure. We first obtained the list of the top 10 brands in each industry from Brand Finance's 2018 rankings.² We then included only those brands that had either U.S.-based or global YouTube brand channels in our sample. As a result of this procedure, we dropped 2 brands – Renault in the automobile industry and Mobile Suit Gundam in the toy industry – even though they were included in the top 10 brands list because they did not have U.S.-based or global YouTube brand channels. Furthermore, we could not include more industries than the 4 industries in our sample mainly because many brands in other industries operate in multiple industries, and it was challenging to match these multi-industry brands with their YouTube brand channels. Most of these multi-industry brands do not have separate YouTube brand channels for the products that they offer in each industry.

The titles of some videos in our sample, mostly among those we collected from global YouTube brand channels, were not in English. To control for language effects, we excluded videos with non-English titles. Furthermore, although we collected data in September 2019 (as of September 2019), we did not include data on branded video content posted in 2019 in our sample to control for age effects and possible position effects. Chowdhury and Makaroff [15] show that most videos on YouTube receive a significant portion of views in the first few days after they are posted.

After all procedures were carried out, we ended up with data for 3,743 videos. The data include the number of subscribers of each YouTube brand channel and, for each video, the number of views and the number of Likes that it received as well as its length, age, and thumbnail URL. Table 1 summarizes the list of brands and the number of videos included in our sample for each industry.

4.2. Variables

As the dependent variable in our model, we use *views* (*V*), which represents the number of views a video received. The main independent variables of interest are the visual attributes of thumbnails that may be associated with the view-through of videos that we discussed in Section 3. These attributes include the *element complexity* (*EC*), *object complexity* (*OC*), *celebrity endorsement* (*CE*), *colorfulness* (*C*), *brightness* (*B*), and *image quality* (*Q*) of thumbnails. We construct these variables by extracting features from thumbnails using three image mining algorithms: Microsoft Azure Computer Vision, Python OpenCV (open-source computer vision), and the Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE). We discuss how we constructed each of these variables

² The brand rankings are available from: <https://brandirectory.com/>.



(a) Objects: {Vehicle, 0.796}, {Person, 0.948};
Tags: {Vehicle, 0.927}, {Land Vehicle, 0.918},
 {Person, 0.824}, {Auto Part, 0.820}, {Man, 0.817},
 {Clothing, 0.790}, {Text, 0.790}, {Wheel, 0.787},
 {Car, 0.134}



(b) Objects: {Person, 0.591}, {Person, 0.751};
Tags: {Toy, 0.999}, {Cartoon, 0.996}, {lego,
 0.973}, {LEGO, 0.965}

Fig. 5. Examples of object detection.

below.

4.2.1. Element complexity

We obtained *EC*, which refers to the number of types of design elements used in a thumbnail, by summing three indicator variables, *product* (*P*), *model* (*M*), and *text* (*T*), which refer to whether a thumbnail includes a product design element, a model design element, and a text design element, respectively.

4.2.2. Product

We obtained the list of objects and tags included in each thumbnail along with the confidence score of each of those objects and tags using two object detection taxonomies – the Detect API and the Tag API of Microsoft Azure Computer Vision – as shown as examples in Fig. 5.³ Using this list, we constructed *P*. *P* is an indicator variable that is set to 1 if the thumbnail includes an industry-specific product (i.e., a product design element) and 0 otherwise. Industry-specific products are products offered by brands in our sample for each industry. These products are chosen from the list of all products detected by the computer vision. It includes “vehicle” and “auto part” for the automobile industry; “drink” and “bottle” for the beverage industry; “food”, “drink”, “menu”, and “kitchen” for the restaurant industry; and “toy”, “LEGO”, “doll”, “gun”, “anime”, and “game” for the toy industry.

4.2.3. Model

Using the list of objects detected for each thumbnail, we constructed *M*, which is set to 1 if the thumbnail includes a person object (i.e., a model design element) and 0 otherwise. As shown as an example in Fig. 5(b), the computer vision detected a toy person or an anime person in a thumbnail as a person object. We manually reviewed all thumbnails identified as having a person object in the toy industry and set *M* to 1 only if the thumbnail included a real person, not a toy person or an anime person.

³ The difference between the Detect API and the Tag API, at a conceptual level, is that the former finds only objects and living things, whereas the latter can also include contextual terms such as indoor (for more details, see: <https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-object-detection>).

4.2.4. Text

Microsoft Azure Computer Vision detects text in an image and reports whether the image has text. Using this, we constructed *T*, which is set to 1 if the thumbnail has text (i.e., a text design element) and 0 otherwise. One challenge we faced while reviewing the computer vision results was that the computer vision detected all types of text, including the text printed on other objects such as license plates, clothes, and packaging as well as logo watermarks in thumbnails, as shown in the examples in Fig. 6. Because text as an independent design element is of interest in this study, we manually reviewed all thumbnails identified as having text and set *T* to 1 only if the text included in a thumbnail was an independent design element (i.e., was not text printed on other objects or a logo watermark).

4.2.5. Object complexity

Using the list of objects detected in each thumbnail and the confidence score of each of those objects, we computed the *OC* of the thumbnail using the following equation suggested by Shin et al. [76]: $OC = -\sum_{i=1}^n p_i \log(p_i)$, where $p_i \in [0, 1]$ is the confidence score of object $i \in \{1, 2, \dots, n\}$ and n is the number of objects included in the thumbnail. A thumbnail that focuses on more objects has a higher value of *OC* than a thumbnail that focuses on fewer objects. For instance, the *OC* of thumbnail (b) ($OC = 2.1296$) in Fig. 3 is greater than that of thumbnail (a) ($OC = 1.4506$).

4.2.6. Celebrity endorsement

Microsoft Azure Computer Vision detects a human face from an image and reports whether it is the face of a celebrity.⁴ Using this, we constructed *CE*, which is set to 1 if a thumbnail includes a celebrity picture and 0 otherwise.

4.2.7. Colorfulness

From each thumbnail, we extracted the *Re*, *Gr*, and *Bl* values, which are the mean values of red, green, and blue pixels in the thumbnail, respectively, using Python OpenCV. We then computed the *C* of each thumbnail using the following equation suggested by Hasler and

⁴ Microsoft Azure Computer Vision recognizes 200,000 celebrities from business, politics, sports, and entertainment (for more details, see: <https://azure.microsoft.com/en-au/blog/cognitive-service-2016-08-25/>).

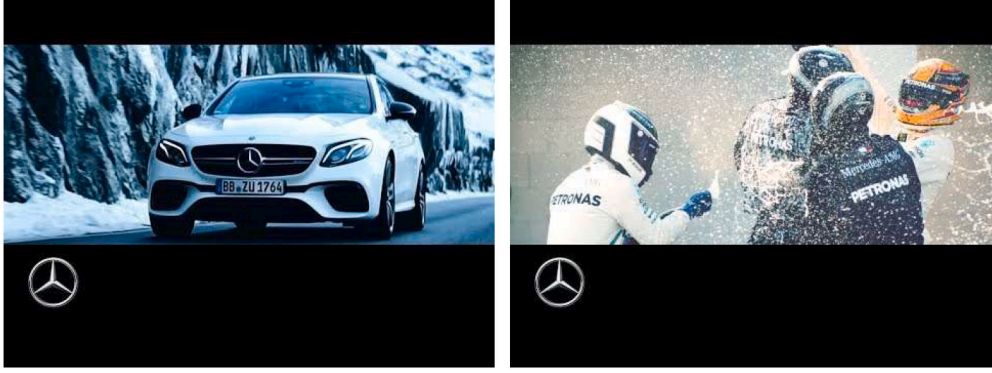


Fig. 6. Examples of thumbnails that have text printed on other objects.

Süsstrunk [33]: $C = \sqrt{\sigma_{rg}^2 + \sigma_{yb}^2} + 0.3\sqrt{\mu_{rg}^2 + \mu_{yb}^2}$, where σ_i and μ_i are the standard deviation and mean of the pixel cloud along direction $i \in \{rg, yb\}$, respectively, where $rg = Re - Gr$ and $yb = 0.5(Re + Gr) - Bl$. Thumbnails that use more colors have higher values of C , as shown in Fig. 7. Thumbnails (c) and (d) in Fig. 7 have higher values of C than thumbnails (a) and (b).

4.2.8. Brightness

Using the Re , Gr , and Bl values extracted from each thumbnail, we computed the B of the thumbnail using the following hue-saturation perceived (HSP) brightness color model suggested by Finley [24]: $B = \sqrt{0.241Re^2 + 0.391Gr^2 + 0.068Bl^2}$. Thumbnails that use brighter colors have higher values of B , as shown in Fig. 7. Thumbnails (b) and (d) in Fig. 7 have higher values of B than thumbnails (a) and (c).

4.2.9. Image quality

We extracted the Q of each thumbnail using the BRISQUE algorithm suggested by Mittal et al. [56]. BRISQUE is a no-reference image quality assessment (IQA) algorithm that measures the degree of distortion of

images. A low value of Q indicates that a thumbnail is distorted, i.e., noisy and blurred. In Fig. 8, thumbnails (a), (b), and (c) have low, moderate, and high values of Q , respectively.

Our independent variables also include the *number of subscribers* (*Subs*) of each YouTube brand channel, the *length* (L) and *age* (A) of each video, and the *number of Likes* (*Likes*) that each video received, all of which have been shown by existing studies to be factors that affect the view-through of videos [6,43,72]. In addition, the relative Google *search volume* (SV) in 2018 is also included as an independent variable to control for the effects of brands' marketing activities outside YouTube on the view-through of videos. The search volume for Ford, which had the highest search volume in 2018 among all brands in our sample, was used as the basis for comparison. The summary statistics of our dependent and independent variables and the correlations of these variables are reported in Tables 2 and 3, respectively.

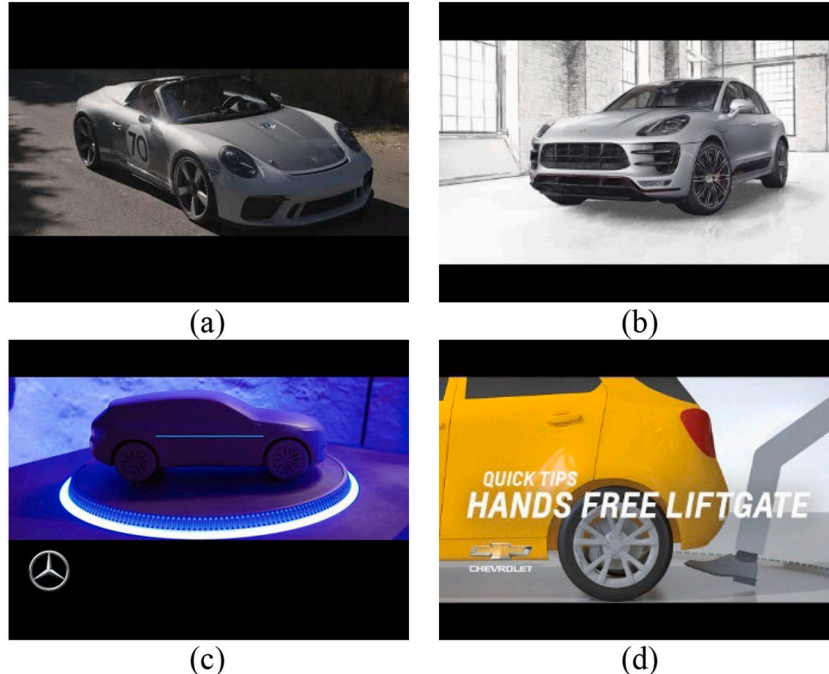


Fig. 7. Thumbnails with (a) low colorfulness and low brightness ($C=7$, $B=40$), (b) low colorfulness and high brightness ($C=4$, $B=137$), (c) high colorfulness and low brightness ($C=84$, $B=42$), and (d) high colorfulness and high brightness ($C=85$, $B=123$)



Fig. 8. Thumbnails with (a) low image quality ($Q = 0.2414$), (b) moderate image quality ($Q = 0.5147$), and (c) high image quality ($Q = 0.8470$).

Table 2
Summary statistics of the variables.

Variable	Mean	S.D.	Min	Max
Views (V)	493,440	1,585,465	84	31,600,000
Element complexity (EC)	1.2012	0.6368	0	3
Object complexity (OC)	0.3903	0.4680	0	4.5293
Celebrity endorsement (CE)	0.0227	0.1490	0	1
Colorfulness (C)	47.39	25.49	0	152.99
Brightness (B)	85.17	32.83	2.15	218.92
Image quality (Q)	0.3423	0.1328	0	1
Subs (unit: millions)	2.9166	3.0190	0.0030	8.7699
Length (L) (unit: minutes)	5.5336	23.28	0.0667	495.05
Age (A) (unit: days)	431	106	243	610
Likes	1,571	5,866	0	165,619
Search Volume (SV)	16.73	22.52	0	89

Note: Approximately 53%, 35%, and 32% of thumbnails in our sample use a product design element, a model design element, and a text design element, respectively.

5. Empirical model and estimation results

5.1. Empirical model

Because our dependent variable, *views* (V), is a count variable and is overdispersed (see Table 2), we use a negative binomial model. The estimation equation of our model is written as follows:

$$Pr(V = v) = \frac{\Gamma(v + \alpha^{-1})}{\Gamma(v + 1)\Gamma(\alpha^{-1})} \left(\frac{1}{1 + \alpha\mu} \right)^{\alpha^{-1}} \left(\frac{\alpha\mu}{1 + \alpha\mu} \right)^v,$$

$$\ln(\mu) = \beta + \gamma \text{INFORMATIVENESS} + \delta \text{AESTHETICS} + \eta \text{OTHER} + \zeta I,$$

where $\Gamma(\cdot)$ is the gamma function, μ is the mean of v , and α is an over-dispersion parameter. $\text{INFORMATIVENESS} \in \{EC, EC^2, OC, OC^2\}$ and $\text{AESTHETICS} \in \{CE, C, C^2, B, B \times C, Q, Q^2\}$ are the main independent variables of interest, and $\text{OTHER} \in \{Subs, L, A, Likes, SV\}$ are other

Table 3
Correlation matrix.

	1	2	3	4	5	6	7	8	9	10	11	12
1. V	1											
2. EC	−0.01	1										
3. OC	0.00	0.17	1									
4. CE	0.01	0.04	0.05	1								
5. C	0.13	0.02	−0.02	0.02	1							
6. B	0.08	0.05	0.04	0.00	0.33	1						
7. Q	−0.09	−0.08	−0.06	−0.03	−0.31	−0.21	1					
8. $Subs$	0.16	0.05	0.10	0.00	0.21	0.24	−0.19	1				
9. L	0.01	0.11	0.05	−0.01	−0.02	0.02	0.08	0.16	1			
10. A	0.03	−0.05	−0.02	0.00	0.05	0.08	−0.03	0.14	0.05	1		
11. $Likes$	0.53	0.04	0.00	0.00	0.02	0.04	−0.03	0.11	0.02	0.00	1	
12. SV	−0.10	0.07	−0.05	0.00	−0.14	−0.15	0.16	−0.30	−0.09	−0.06	−0.11	1

Note: Bold values indicate that the correlation is significant at the 5% level.

independent variables. Following He et al. [35], to control for industry fixed effects, we include industry dummies, $I \in \{I_1, I_2, I_3\}$, in the equation.

We include the squared terms of EC , OC , C , and Q in the equation to control for any possible nonmonotonic relations of these variables with the view-through videos. As discussed in Section 3, the findings of earlier studies are mixed and suggest that the effects of the visual complexity – element complexity and object complexity – of a design on its effectiveness in conveying information and the effects of the colorfulness and image quality of a design on its aesthetics may not be monotonic. We also include the interaction term $C \times B$ in the equation to control for a possible interaction effect between the two color-related properties: colorfulness and brightness.

5.2. Estimation results

The estimation results of our model are reported in Table 4 (in column (2)). The estimation results of the model without the squared terms and the interaction term are also reported in Table 4 (in column (1)) for reference. Our main independent variables – *INFORMATIVENESS* and *AESTHETICS* – are all significant, suggesting that, as argued by industry experts [10,26], the visual attributes of thumbnails are indeed associated with the view-through of videos. This may suggest that the visual attributes of thumbnails persuade potential viewers to click and watch videos via the central route and peripheral route.

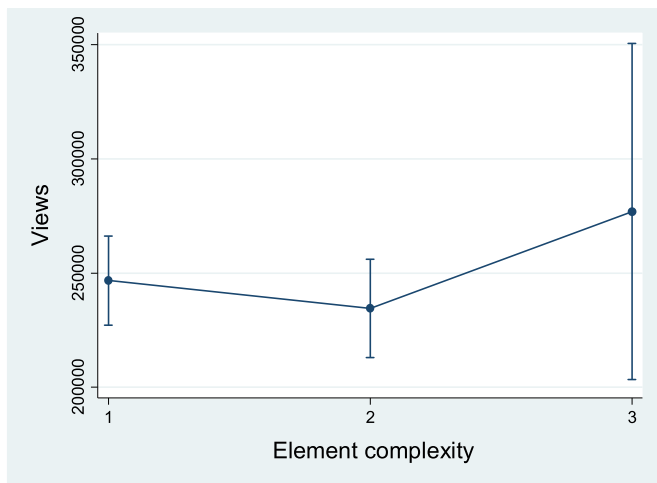
The coefficients of *INFORMATIVENESS* suggest that although the EC of thumbnails and the view-through of videos have a U-shaped relation, the OC of thumbnails and the view-through of videos have an inverted U-shaped relation (see Fig. 9). These results suggest that having all three types of design elements in a thumbnail may help convey information; however, focusing on too many objects can hinder the processing of the information.

The coefficients of *AESTHETICS* suggest the following results. First, the relation between the CE of thumbnails and the view-through of videos is positive. This may suggest that compared to thumbnails

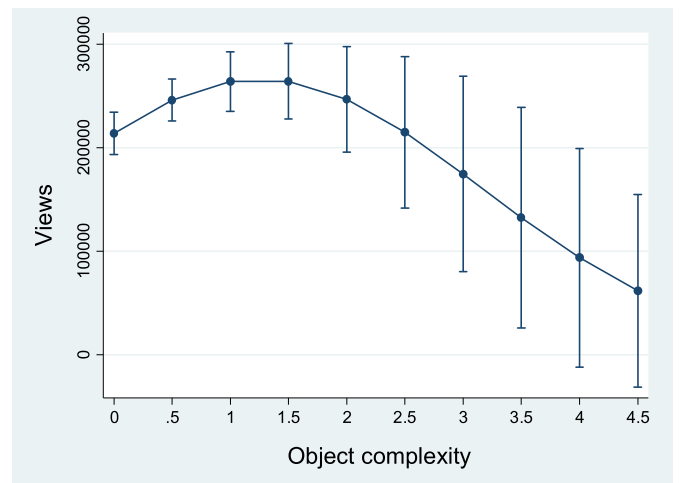
Table 4
Estimation results.

Industry	(1) Overall	(2)	(3) Automobile	(4) Beverage	(5) Restaurant	(6) Toy
<i>EC</i>	−0.0976** (0.0386)	−0.3775*** (0.1165)	−0.5509*** (0.2062)	−0.2990 (0.3559)	1.7789* (0.9661)	−0.3644** (0.1658)
<i>EC</i> ²		0.1088** (0.0428)	0.2438*** (0.0649)	0.0396 (0.1386)	−0.0233 (0.4268)	0.1691** (0.0685)
<i>OC</i>	0.1208* (0.0569)	0.3484*** (0.1162)	0.4334** (0.1836)	−0.5557 (0.3648)	−3.9777** (1.6083)	0.4155*** (0.1491)
<i>OC</i> ²		−0.1387** (0.0582)	−0.2673*** (0.1000)	0.3589** (0.1814)	2.4853*** (0.8041)	−0.1363** (0.0673)
<i>CE</i>	1.0025*** (0.1717)	1.0291*** (0.1721)	0.9752*** (0.2613)	1.7239*** (0.3171)	0.1572 (0.9363)	0.1027 (0.2747)
<i>C</i>	0.0025** (0.0012)	−0.0198*** (0.0042)	0.0238*** (0.0061)	−0.0026*** (0.0162)	0.0157 (0.0403)	0.0162** (0.0071)
<i>C</i> ²		0.0001*** (0.0000)	−0.0003*** (0.0001)	0.0003** (0.0001)	−0.0001 (0.0003)	0.0000 (0.0000)
<i>B</i>	−0.0033*** (0.0009)	−0.0077*** (0.0016)	−0.0019 (0.0021)	−0.0188*** (0.0044)	−0.0108 (0.0135)	0.0058** (0.0026)
<i>C × B</i>		0.0001*** (0.0000)	−0.0001 (0.0001)	0.0004*** (0.0001)	0.0001 (0.0003)	−0.0001* (0.0001)
<i>Q</i>	−0.0601 (0.2370)	2.9586*** (0.8119)	4.8393*** (1.1694)	8.9840*** (2.1522)	−1.0698 (5.0008)	3.7176*** (1.2711)
<i>Q</i> ²		−4.2264*** (0.9778)	−6.2007*** (1.2305)	−9.5619*** (2.6407)	7.2676 (6.7720)	−7.1721*** (1.7360)
<i>Subs</i>	0.0693*** (0.0123)	0.0734*** (0.0123)	0.5305*** (0.0909)	0.0021 (0.0260)	1.7578 (2.2470)	0.1662*** (0.0295)
<i>L</i>	−0.0027*** (0.0009)	−0.0021** (0.0009)	−0.0300*** (0.0055)	0.0032** (0.0014)	−0.1552 (0.0999)	−0.0045* (0.0024)
<i>A</i>	−0.0004 (0.0002)	−0.0003 (0.0002)	−0.0005 (0.0003)	−0.0003 (0.0007)	−0.0059*** (0.0019)	0.0001 (0.0004)
<i>Likes</i>	0.0002*** (0.0000)	0.0002*** (0.0000)	0.0004*** (0.0000)	0.0001*** (0.0000)	0.0002*** (0.0001)	0.0002*** (0.0000)
<i>SV</i>	−0.0058*** (0.0015)	−0.0061*** (0.0015)	−0.0047*** (0.0017)	0.0937*** (0.0359)	0.0142*** (0.0166)	−0.0246* (0.0134)
<i>I</i> ₁	0.3469*** (0.1091)	0.3365*** (0.1105)				
<i>I</i> ₂	2.2628*** (0.1444)	2.2872*** (0.1433)				
<i>I</i> ₃	1.4760*** (0.1025)	1.5296*** (0.1031)				
<i>Constant</i>	11.6014*** (0.1766)	11.8087*** (0.2536)	10.0849*** (0.3388)	12.1130*** (0.6350)	12.1240*** (2.5627)	10.8786*** (0.4311)
<i>Observations</i>	3,743	3,743	1,314	429	130	1,870
<i>Pseudo R</i> ²	0.0188	0.0193	0.0211	0.0307	0.0257	0.0105

Note: Standard errors are in parentheses; * $p \leq 0.1$, ** $p \leq 0.05$, *** $p \leq 0.01$.



(a)



(b)

Fig. 9. The relations between (a) the element complexity of thumbnails and the view-through of videos and (b) the object complexity of thumbnails and the view-through of videos.

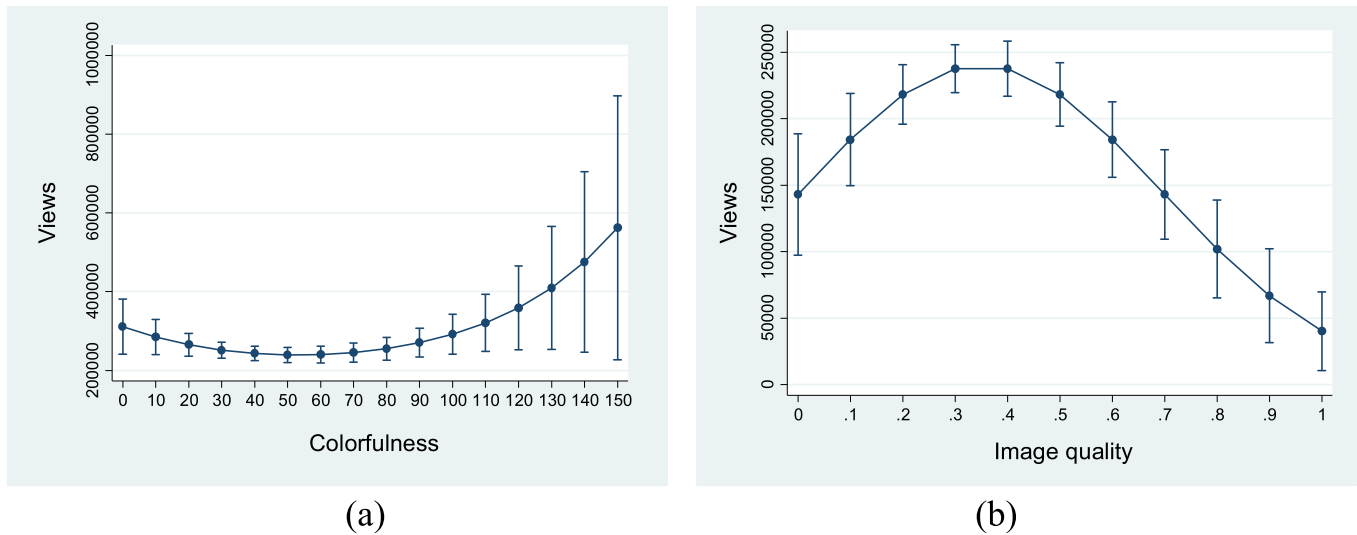


Fig. 10. The relations between (a) the colorfulness of thumbnails and the view-through of videos and (b) the image quality of thumbnails and the view-through of videos.

without a celebrity picture, thumbnails with a celebrity picture are more visually appealing and thus persuade potential viewers to click and watch videos via the peripheral route. Second, the C of thumbnails and the view-through of videos have a U-shaped relation (see Fig. 10(a)). Furthermore, the relation between the B of thumbnails and the view-through of videos is negative if the colorfulness of thumbnails is low, and it is positive if the colorfulness of thumbnails is high (see Fig. 11). These results suggest that thumbnails with a combination of low colorfulness and low brightness or high colorfulness and high brightness may be more effective in persuading potential viewers to click and watch videos via the peripheral route. These results provide additional insight into the findings of earlier studies showing that brighter images are more visually appealing [51,83]. Finally, the Q of thumbnails and the view-through of videos have an inverted U-shaped relation (see Fig. 10(b)). This may suggest that neither a distorted nor an over-retouched thumbnail image is effective in persuading potential viewers to click and watch videos via the peripheral route. As thumbnail images taken from videos are often distorted, industry experts suggest that enhancing the quality of thumbnail images is important for obtaining more video views [61,86]. However, as discussed in Section 3, retouching advertisement images may not necessarily improve the

effectiveness of advertisements [17,88].

The results of our other independent variables are consistent with our expectations and the findings of earlier studies [6,43,72]. *Subs* and *Likes* are positively associated, and *L* is negatively associated with the view-through of videos. *SV* is also negatively associated with the view-through of videos, suggesting that there may be a substitute relation between brands' marketing activities within and outside of YouTube. *A* is estimated to be not significant. The reason may be that, as discussed in Section 4, the age effects in our data are already controlled to some degree, as we did not include the data on videos posted in 2019 in our sample.

5.3. Industry-level analysis

It has been shown in the literature that individuals' reactions to aesthetics may be different across different product groups [31]. To test whether the relation between the visual attributes of thumbnails and the view-through of videos is different across different industries, in this section, we run the model separately for each industry. The estimation results are reported in Table 4 (in columns (3)–(6)). Perhaps because the observations for the beverage and restaurant industries are not sufficiently large, the estimation results for these two industries are less significant. On the other hand, the estimation results for the automobile and toy industries are fairly consistent with the estimation results for industries overall for most variables, except *CE*, *C*, and *B*. The estimation results for these three variables suggest the following findings. First, whereas there is a significant celebrity endorsement effect in the automobile industry, the celebrity endorsement effect is not significant in the toy industry. This finding may suggest that a celebrity is not an important factor for children, who are the main viewers of videos in the toy industry, when they decide which videos to click and watch. Second, whereas thumbnails with high colorfulness and low brightness may be more effective in persuading viewers to click and watch videos in the toy industry, thumbnails with low colorfulness may be more effective in the automobile industry. Brightness is not a significant factor in the automobile industry.

6. The titles of videos

Viewers may decide which videos to click and watch based on the information they obtain from the titles, in addition to the thumbnails, of videos when they browse through an online video platform. In fact,

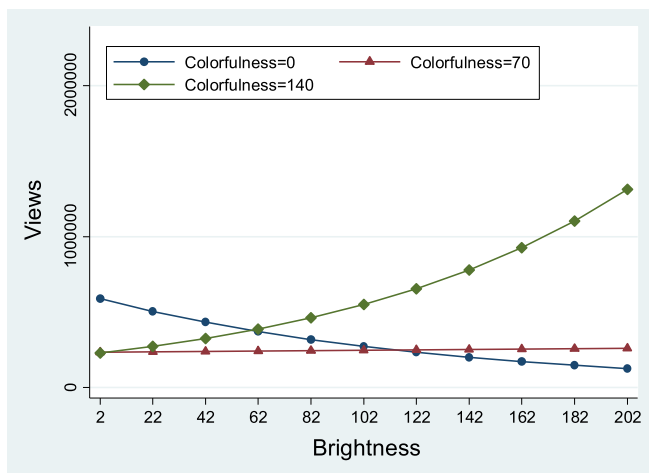


Fig. 11. The relations between the brightness of thumbnails and the view-through of videos for different levels of colorfulness.

Table 5

Estimation results of the extended models with additional variables.

	(1)		(2)		(3)		(4)	
	Est.	Std. Err.	Est.	Std. Err.	Est.	Std. Err.	Est.	Std. Err.
<i>EC</i>	-0.3775***	0.1165	-0.3729***	0.1163	-0.2685**	0.1154	-0.3773***	0.1164
<i>EC</i> ²	0.1088**	0.0428	0.1048**	0.0428	0.0798*	0.0420	0.1056**	0.0430
<i>OC</i>	0.3484***	0.1162	0.3133***	0.1136	0.2781**	0.1140	0.3049***	0.1156
<i>OC</i> ²	-0.1387**	0.0582	-0.1122**	0.0560	-0.1158**	0.0564	-0.1234**	0.0573
<i>CE</i>	1.0291***	0.1721	1.0112***	0.1717	1.1479***	0.1704	0.9627***	0.1730
<i>C</i>	-0.0198***	0.0042	-0.0191***	0.0042	-0.0204***	0.0042	-0.0206***	0.0043
<i>C</i> ²	0.0001***	0.0000	0.0001***	0.0000	0.0001***	0.0000	0.0001***	0.0000
<i>B</i>	-0.0077***	0.0016	-0.0076***	0.0016	-0.0087***	0.0016	-0.0075***	0.0016
<i>C × B</i>	0.0001***	0.0000	0.0001***	0.0000	0.0001***	0.0000	0.0001***	0.0000
<i>Q</i>	2.9586***	0.8119	2.9792***	0.8078	2.5205***	0.8086	3.0127***	0.8099
<i>Q</i> ²	-4.2264***	0.9778	-4.2813***	0.9720	-3.7147***	0.9739	-4.3887***	0.9745
<i>Subs</i>	0.0734***	0.0123	0.0672***	0.0123	0.0962***	0.0118	0.0734***	0.0122
<i>L</i>	-0.0021**	0.0009	-0.0020**	0.0009	-0.0021**	0.0010	-0.0016*	0.0010
<i>A</i>	-0.0003	0.0002	-0.0003	0.0002	-0.0003	0.0002	-0.0003	0.0002
<i>Likes</i>	0.0002**	0.0000	0.0002**	0.0000	0.0002**	0.0000	0.0002**	0.0000
<i>SV</i>	-0.0061***	0.0015	-0.0062***	0.0015	-0.0045***	0.0015	-0.0060***	0.0015
<i>TC</i>			-0.2050***	0.0478				
<i>Category</i>					0.8638***	0.0872		
<i>Sentiment_p</i>							0.2185***	0.0582
<i>Sentiment_N</i>							0.3468***	0.0795
<i>I₁</i>	0.3365***	0.1105	0.3414***	0.1103	0.3069***	0.1076	0.2602**	0.1113
<i>I₂</i>	2.2872***	0.1433	2.2711***	0.1429	2.3075***	0.1416	2.3380***	0.1434
<i>I₃</i>	1.5296***	0.1031	1.6154***	0.1049	1.2239***	0.1052	1.4148***	0.1059
<i>Constant</i>	11.81***	0.2536	11.89***	0.2251	11.79***	0.2511	11.75***	0.2532
<i>Observations</i>	3743		3743		3743		3743	
<i>Pseudo R²</i>	0.0193		0.0195		0.0204		0.0207	

Note: * $p \leq 0.1$, ** $p \leq 0.05$, *** $p \leq 0.01$.

although video titles are, in often cases, truncated after a fairly small number of words when they are shown in an online video platform, anecdotal evidence suggests that compelling video titles improve the view-through of videos [3]. To control for this possible effect of video titles on the view-through of videos, we extend our model by including the following additional variables.

First, following the literature suggesting that the topic complexity of the textual content of advertisements may be a central route cue [76], we construct a variable, the *topic complexity (TC)* of video titles, and include it in our model. We construct *TC* using the latent Dirichlet allocation (*LDA*) topic modeling approach suggested by Shin et al. [76]. After obtaining the optimal number of topics of 40 using the topic coherence score, we convert each video title into a topic distribution vector and obtain *TC* using the following equation: $TC = -\sum_{i=1}^t d_i \log(d_i)$, where $d_i \in [0, 1]$ is the weight of topic $i \in \{1, 2, \dots, t\}$ and t is the number of topics included in the title. Video titles with more topics have a higher value of *TC* than video titles with fewer topics.

Second, we include a video cluster dummy, *Cluster*, in our model by clustering video titles using the term frequency-inverse document frequency (TF-IDF) bag-of-words and K-means algorithm. Using the elbow method, we categorize video titles into two clusters. The top five keywords of the first cluster are “trailer”, “world”, “new”, “game”, and “episode”, and those of the second cluster are “official”, “teaser”, “spot”, “season”, and “game”.

Finally, after obtaining the sentiment polarity of each title using the VADER (Valence Aware Dictionary and sEntiment Reasoner) algorithm, we construct the title sentiment dummies – *Sentiment_p* and *Sentiment_N* – and include them in our model. VADER is a lexicon- and rule-based algorithm for sentiment analysis proposed by Hutto and Gilbert [38]. If the VADER compound score is positive, the title is considered to have positive sentiment, and *Sentiment_p* is set to 1. If the VADER compound score is negative, the title is considered to have negative sentiment, and *Sentiment_N* is set to 1. If the VADER compound score is zero, the title is considered to have neutral sentiment, and both *Sentiment_p* and *Sentiment_N* are set to zero. Fifty-six percent of videos in our sample have titles with neutral sentiments, and 30% and 14% of videos have titles with positive and negative sentiments, respectively.

The estimation results of the extended models with these additional variables are reported in Table 5 (in columns (2), (3), and (4)). We also report the estimation results of our base model in Table 5 (in column (1)) for reference. These results suggest that, first, our findings on the relation between the visual attributes of thumbnails and the view-through of videos that we discussed in Section 5 may be quite robust. Furthermore, the *TC* is negatively associated with the view-through of videos, which may suggest that less complex video titles are more efficient in conveying information. This may be plausible, as potential viewers can see only a fairly small number of words in video titles when they browse through an online video platform. Last, both *Sentiment_p* and *Sentiment_N* are positively associated with the view-through of videos. This may suggest that video titles with positive sentiments (i.e., that include positive words such as “great” and “enjoy”) and negative sentiments (i.e., that include negative words such as “disaster” and “insane”) are more effective in persuading potential viewers to click and watch videos than titles with neutral sentiments.

7. Conclusion

Using 3,745 marketing videos posted by 38 top brands across 4 industries – the automobile, beverage, restaurant, and toy industries – on YouTube, we investigated the relation between the visual attributes of thumbnails and the view-through of videos. We found that the visual attributes of thumbnails are associated with the view-through of videos, which may suggest that the visual attributes of thumbnails persuade potential viewers to click and watch videos via the central route and peripheral route. In particular, having more types of design elements in a thumbnail may help convey information; however, focusing on too many objects can hinder the processing of the information. Furthermore, thumbnails with a celebrity picture, with a combination of high colorfulness and high brightness or low colorfulness and low brightness, and with moderate image quality may be more effective in persuading potential viewers to click and watch videos via the peripheral route. The industry-level analysis reveals that potential viewers' reactions to the aesthetics of thumbnails may be different across different industries.

Our study makes the following contributions. First, our study

enhances our understanding of the role of visual attributes of images in persuading viewers in the online environment. Furthermore, complementing a growing body of literature studying the factors that affect the view-through of videos, our study shows the relation between the visual attributes of thumbnails and the view-through of videos. This may provide a possible theoretical ground for studies on information processing and decision support that focus on automation thumbnail selection. Finally, our study provides useful and practical guidelines for designing templates for optimal thumbnails that grab potential viewers' attention and yield more video views.

Nevertheless, our study is not without limitations. First, the literature has shown that YouTube recommendations are an important factor affecting the view-through of videos [89]. However, because a dataset that captures YouTube recommendations across the board is not readily available, as this process is heavily personalized [54], our model controls for recommendation effects only indirectly through measures such as the number of Likes that videos received and the number of subscribers of each channel. How the YouTube recommendation algorithm works is mostly unknown, but articles have suggested that YouTube considers factors such as the popularity of channels and how much people like videos when it decides which videos to recommend [20,39,54]. Future research can investigate the effect of recommendations on the relation between the visual attributes of thumbnails and the view-through of videos using a more direct measure. Second, our study focuses on the relation between the visual attributes of thumbnails and the view-through of videos when viewers browse through an online video platform, and it does not model viewers' searches. Although some viewers may begin browsing by searching, Cunningham and Nichols [18] show that viewers eventually find relevant videos using browsing, and viewers report that browsing is a useful and efficient way to find relevant videos. Nevertheless, one may want to extend our study and investigate how viewers' searching moderates the relation between the visual attributes of thumbnails and the view-through of videos. Third, our data can be extended further. Our dataset includes data on 3,745 marketing videos posted by 38 top brands across 4 industries. It would be worthwhile to collect data from more brands covering more industries and to further generalize the results. In particular, given that "how-to" videos are growing rapidly in popularity [57], examining the relation between the visual attributes of thumbnails and the view-through of videos, particularly for "how-to" videos, would be valuable. Finally, although our findings are based on grounded theory, they are derived from regression models and may not be sufficient to make causal claims. It would be valuable to extend our model and investigate the relation between the visual attributes of thumbnails and the view-through of videos based on causal models and establish causality.

CRedit authorship contribution statement

Byungwan Koh: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Fuquan Cui:** Software, Data curation, Formal analysis, Writing – original draft.

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